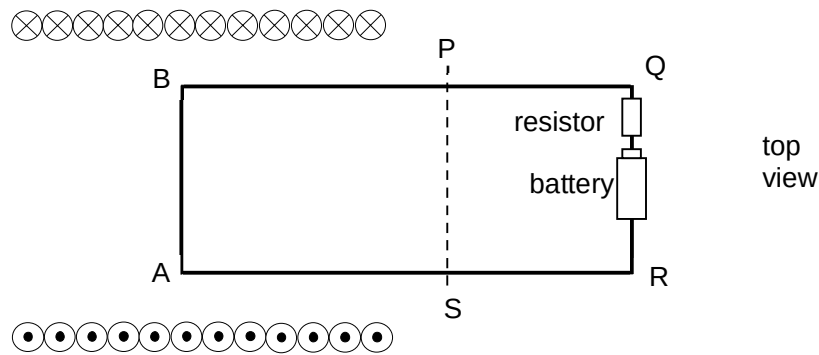


- 24 The figure below shows the top view of a current balance where the rigid rectangular wire loop ABQR pivoted at PS is in equilibrium. It is connected in series with an ideal 2.0 V battery and a $0.025\ \Omega$ resistor of a total mass of 300 g. Part of the wire loop is placed inside a solenoid. The mass of the loop can be taken to be negligible and the wire has no resistance.



The length of the side AB is 6.0 cm and $SR = \frac{2}{5} ASR$.

What is the magnitude of the magnetic flux density in the solenoid?

- A** 0.37 T **B** 0.41 T **C** 3.7 T **D** 4.1 T

Ans: B

By principle of moments,

$$F_B \times AS = mg \times SR$$

$$B \times \frac{2}{0.025} \times 0.060 \times 3L = 0.300 \times 9.81 \times 2L$$

$$B = 0.41\text{T}$$

Distractors:

Never convert units – 4.1 T

Use wrong distance – $AS = 5L$ and $SR = 3L$ – 0.37 T

Use wrong distance and never convert units = 3.7 T

- 25 When a light bulb is connected across an a.c. source of peak voltage 150 V, the mean power dissipated is 13 W. Two such light bulbs are now connected in series to the electrical mains of 240 V r.m.s.

What is the peak voltage across each light bulb and the total power dissipated in the light bulbs?

	peak voltage across each light bulb / V	total power dissipated in the light bulbs / W
A	120	17
B	170	33
C	120	67
D	170	17

Ans: B

$$V_{rms} = \frac{150}{\sqrt{2}}$$

$$P = \frac{V_{rms}^2}{R} \Rightarrow R = 865\Omega$$

$$R_{total} = 865 \times 2$$

$$P_{total} = \frac{240^2}{865 \times 2} = 33$$

$$V_0 \text{ across each bulb} = 120\sqrt{2} = 170$$

Distractors:

For 17 W, take peak voltage as $V_{r.m.s.}$

For 67 W, assume only one bulb connected to 240 V rms source.

26. A voltmeter reads 80 V when measuring the potential difference across a load of $10\ \Omega$.